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Computer Engineering Student

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Profile

Computer Engineering student at Marmara University, Co-founder & CTO of **Giyiver** (a clothing-swap startup backed by investment received through Shell ILERIDE) and AI Engineering Intern at Orion Innovation, building LLM, vision-language, retrieval and speech AI systems. Solid experience in Python, machine learning and model evaluation from industry work, the Google AI and Technology Academy and personal projects. I enjoy turning models into reliable, measurable products.

Education

2023 - 2027 **B.Sc. in Computer Engineering**, *Marmara University*, Istanbul
Third-year student

2019 - 2023 **High School**, *Bahçelievler Anatolian High School (German)*

Experience

June 2026 - Present **Co-founder & CTO**, ***Giyiver***, Istanbul, giyiverr.com

Turkish clothing-swap marketplace (barter, donation, points) live at giyiverr.com. Backed by investment received through Shell ILERIDE. We are a team of 7, and I am responsible for the entire technical side of the project: web, mobile, backend, infrastructure, AI, security and testing.

- **AI listing autofill:** a vision-language model (NVIDIA NIM, **Nemotron 3.5 Super VL**) turns uploaded photos into title, category, colour, condition and description; images are downscaled client-side and sent as validated base64, so nothing is stored before publishing.
- **Model selection and hedged inference:** benchmarked candidate VLMs on real listing photos, dropped unreliable ones, disabled reasoning mode for **2-10 s** typical latency, and added a fallback model that races the primary after 12 s (first valid answer wins, the loser is cancelled).
- **Multimodal moderation:** one VLM call checks all photos and text of a listing (real clothing, same item in every photo, consistent description); free text also passes a local profanity filter and an AI check, built to fail open and backed by database triggers.
- **Verified-image tokens:** HMAC-signed, user-bound tokens carry SHA-256 digests of the analysed images; at publish time the server re-hashes the uploaded files and skips the second VLM call only if every digest matches, saving latency and cost without trusting the client.
- **Platform:** Next.js on Cloudflare Pages with edge functions, Supabase Postgres with row-level security (**57** migrations) and server-side points escrow, an Expo/React Native app, crawler-only rendering for SEO, and **570+** automated tests including SQL tests on a real Postgres engine.

July 2026 - Present

AI Engineering Intern, *Orion Innovation*, Istanbul

Building LLM, retrieval and speech systems across three projects: **SOSIS** (Software product line Optimization for Safety-/mission-critical Industrial Systems, ITEA 4), an AI-driven test management platform; **GENIUS** (Generative AI for the Software Development Life Cycle), LLM-based requirement extraction; and **CLEAR** (Call Listener Emergency Audio Recognition), real-time audio intelligence for emergency call centers.

- **SOSIS:** Designed a hybrid retrieval pipeline (dense embeddings + BM25, reciprocal rank fusion, GraphRAG) with optional listwise **LLM reranking** that links requirements, tests and source code automatically.
- Benchmarked **20+ LLMs** across 5 providers and **10+ embedding models** on hand-labeled gold sets with paired bootstrap confidence intervals, reaching **MRR 0.71** (requirement-to-test) and **Hit@5 97.5%** (test-to-code); wrote the study up as a research paper.
- Built an LLM pipeline that turns requirements into runnable Playwright tests, grounded with retrieved code, a graph-based page selector (page@3 **63%→90%**) and a headless-browser probe of the live UI, plus checks that feed the model its own mistakes back before the first run.
- Built LLM agents that adapt tests to a product variant, run them, repair failures from the run report and review the outcome (LLM-as-a-judge); mapped requirements to variants with a hybrid embedding scorer (**F1 0.74**).
- Unified OpenRouter, NVIDIA NIM, Groq and Google AI Studio behind one client with provider routing, price caps and a token-per-minute scheduler.
- **GENIUS:** Optimized the multi-stage LLM requirement-extraction pipeline by batching zone classification, query expansion and rationale generation: LLM requests per document fell from **95 to 9** (zone classification **107 s→3 s**, now well inside Groq's rate limit), and controlling reasoning effort on GLM 5.3 Flash halved token use (**336K→165K**).
- Evaluated the extracted requirements against a hand-written ground truth (**12 of 18** items matched fully, 6 partially, none missed), and integrated GENIUS into SOSIS so that its EARS/LESS requirements and labels (ISO 25010 quality dimension, actor, classification) feed test generation and variant mapping.
- **CLEAR:** Built a streaming pipeline that runs speech recognition (Qwen3-ASR with LoRA), speaker diarization, speech emotion, stress and audio-event models on live 911/112 calls, fully on-premise on CPU.
- Trained calibrated accent classifiers on Whisper encoder features and combined emotion and stress signals for distress detection (**AUROC 0.87** vs 0.71 for stress alone).
- Evaluated models on public benchmarks: **6.3%** WER on clean speech, **86%** emotion accuracy (vs **41%** for the earlier approach), first words appear on screen in about **2 s**.

January 2026 - Present

AI Team Lead, *GDG - Google Developer Group Marmara*

- Lead the AI team at GDG Marmara, driving technical projects and developer-focused community activities.
- Work with the team on project planning, product thinking and delivery for software projects and community events.
- Help turn ideas into buildable projects while building coordination, ownership and community skills.

February 2026 - June 2026

ILERIDE Program | Trainee & Startup Founder, *Shell*, Istanbul

- Took part in Shell's ILERIDE program, which focused on structured thinking, problem framing, communication and business skills.
- **Giyiver**, the startup I co-founded and lead as CTO, **received investment** through the program.
- Used what I learned in the program for Giyiver's product and technical decisions, and to explain technical work to non-technical people.

December 2024 - August 2025

Data Science Trainee, *Google AI and Technology Academy*, Istanbul

- Completed an intensive 8-month program in applied data science, machine learning and end-to-end project development.
- Learned to work with real datasets, from analysis and statistics to model evaluation and turning results into insights, through coursework and team projects.
- Took part in datathons and bootcamps, building models and working in teams under tight deadlines.
- Helped build an AI-supported mobile app for preliminary dementia assessment during the bootcamp, finishing **3rd among 205 teams**.

Projects

2026 **Credit Card Fraud Detection System**, *Personal Project*

- Developed a production-ready machine learning pipeline for fraud detection using a dataset of **1.04M+ transactions**.
- Implemented **leakage-free feature engineering** using point-in-time calculations (velocity, behavioral patterns) to prevent future data leakage.
- Optimized and compared LightGBM, XGBoost, CatBoost, and Random Forest models, achieving a **0.925 PR-AUC** and **0.85 F1-Score**.
- Addressed extreme class imbalance (**0.58% fraud**) by prioritizing Precision-Recall AUC optimization to minimize false positives in a financial context.
- Built automated diagnostic tools for data distribution analysis and model performance visualization using Pandas, Matplotlib, and Seaborn.
- GitHub Repository

2026 **Unsolved: AI-Driven Problem Discovery Platform & Automated Scraper**, *Personal Project*

- Built an automated research pipeline with **Next.js 16** and **TypeScript** that scrapes real-time customer signals from **Reddit API** and **App Store RSS** feeds to identify unaddressed needs and market gaps.
- Developed a heuristic scoring engine that analyzes raw text signals using **keyword-based NLP** and sentiment weights to quantify **Pain Intensity**, **Frequency**, and **Willingness to Pay (WTP)**.
- Built an ingestion layer with **deduplication logic** and **automatic categorization** (Bug, Feature Request, Experience Gap), transforming thousands of raw social signals into actionable product insights.
- Added a persistence layer with **Prisma ORM** and **PostgreSQL**, featuring **upsert mechanisms** to maintain a synchronized "Problem Store" that tracks market trends and competitor gaps over time.
- Integrated an **AI Analysis Engine** that utilizes custom prompt versioning to generate structured product strategy reports, including MVP steps, solution ideas, and risk assessments based on ingested market signals.
- GitHub Repository

2026 **TaskFlow Collaborative Kanban Platform**, *Personal Project*

- Built a full-stack Kanban board application with **Next.js**, **React**, **TypeScript**, **Prisma**, and **PostgreSQL**, focused on practical team task management.
- Developed responsive dashboard and landing page flows, including drag-and-drop board interactions, due dates, multi-assignee task management, and mobile-friendly board navigation.
- Implemented **role-based collaboration** features such as board sharing, viewer/editor permissions, self-leave flows for invited users, and access-aware UI/route protection.
- Designed and maintained API routes, database schema migrations, and state management for real-time-feeling collaborative workflows across shared boards.
- GitHub Repository

2025 **Crypto Analysis and Trading Bot**, *Personal Project*

- Built a Python trading bot based on custom quantitative strategies.
- Designed an architecture that screens **200+ crypto assets** in about **2 minutes** and reached a **66% success rate** with a **1:2.5 risk-reward ratio** in a real-time Telegram signal system.
- Evaluated LightGBM, CatBoost, and TabNet during R&D, then refactored the live system into a custom strategy-based architecture to reduce latency.
- Processed high-volume market data with NumPy and Pandas and maintained trade logs and historical records in MongoDB.
- Automated 24/7 trade lifecycle tracking through an additional "Crypto Signal Tracker" bot.

2025 **Datathon Competition**, *Google AI and Technology Academy*

- Led the data analysis and visualization of a **210,000-row dataset** in a **4-person cross-functional team**.
- Built price prediction models for a new **56,000-row dataset** using LightGBM, Random Forest, and XGBoost.
- Interpreted model outputs and translated complex findings into actionable visual insights with Matplotlib and Seaborn.

2025 **AI-Integrated Health Application**, *Bootcamp Project*

- Developed a mobile application with a **5-person team** for the preliminary assessment of cognitive conditions such as hyperactivity, dementia, and autism.
- Developed the Python backend and core Dart frontend components, leading the team to rank in the **top 3 among 205 competing teams**.
- Architected a voice pronunciation test module in Python, managed Firebase operations, and integrated systems through REST APIs.
- Optimized several Dart screens and fixed critical security vulnerabilities.
- GitHub Repository

2025 **Fire Detection System for Live Security Cameras**, *Computer Vision*

- Trained a model on a dataset of **95,000 images** using YOLO 11.
- Reached **82.8%** smoke detection accuracy and **79.6%** fire detection accuracy, with especially strong results on black-and-white images.
- Tested the system on both live camera streams and recorded video footage, marking detected regions with bounding boxes.
- GitHub Repository

2025 **Real-Time Chat Application**, *Team Project*

- Built a real-time messaging application with a 3-person team using HTML, CSS (Tailwind), JavaScript, React, Node.js, MongoDB, and Socket.io.
- Managed Node.js backend tasks, real-time connection management with Socket.io, MongoDB workflows, and media storage with Cloudinary.

2024 **Automated Linux Backup System**, *Personal Project*

- Engineered a modular **Bash-based backup system** with support for both interactive TUI and fully automated headless execution modes.
- Integrated a dynamic terminal-based file explorer and user interface using the **dialog** library to enhance user experience.
- Ensured data security through **atomic scheduling via crontab**, recursive file management with find, and optimized archive compression using tar.
- Implemented a centralized logging mechanism to provide end-to-end traceability and auditability of backup operations.

2024 **Energy Distribution and Fault Response Simulation**, *Team Project*

- Built a city-scale energy fault simulation using Spring Boot and JavaScript to model real-time detection and response workflows for urban power outages.
- Designed the full JavaScript/HTML interface and supported backend integration for optimized outage-response workflows.

2024 **Personal Series/Movie Watchlist Platform**, *Team Project*

- Developed a social watchlist application using Python (backend) and CustomTkinter (frontend) within a 3-person team.
- Engineered core backend features and designed the RESTful API layer to ensure seamless communication between system components.
- End-to-end feature development including personal watchlist management and social interaction modules.

Technical Skills

Programming Languages	Python, TypeScript, JavaScript, Java, C/C++, Dart, Bash

LLM Engineering	LLM agents, LLM-as-a-judge, structured output, prompt engineering, reasoning-effort control, request batching, multi-provider routing and rate-limit scheduling (OpenRouter, NVIDIA NIM, Groq, Google AI Studio)
Retrieval and RAG	Hybrid search (BM25 + dense embeddings), score fusion (RRF, DBSF), GraphRAG, listwise LLM reranking, query expansion, sentence-transformers, Qdrant
Multimodal and Vision	Vision-language models (Nemotron VL), multimodal content moderation, VLM benchmarking and fallback inference, YOLO object detection
Speech and Audio	Speech recognition (Qwen3-ASR, Whisper), LoRA fine-tuning, speaker diarization, speech emotion recognition (wav2vec 2.0, emotion2vec), stress and audio-event detection, real-time streaming, on-premise CPU inference
Machine Learning	PyTorch, scikit-learn, LightGBM, XGBoost, CatBoost, TabNet, Keras, Pandas, NumPy, Matplotlib, Seaborn
Model Evaluation	Hand-labeled gold sets, MRR, Hit@k, paired bootstrap confidence intervals, WER, AUROC, PR-AUC, F1, probability calibration
Web and Mobile	FastAPI, Next.js, React, Node.js, Expo / React Native, Spring Boot, REST, WebSocket, Socket.io
Data and DevOps	PostgreSQL, Supabase, MongoDB, Prisma, Firebase, SQL / MSSQL, Docker, Cloudflare Pages, GitLab CI/CD, Playwright, pytest, Git, uv

Languages

Turkish	Native
English	B2
German	A2-B1

Certificates

2026	Artificial Intelligence Specialization Program - Foundation Training Certificate (Republic of Türkiye Ministry of Industry and Technology)
2026	Introduction to Open Source License Compliance Management (The Linux Foundation)
2026	Model Context Protocol: Advanced Topics (Anthropic)
2026	AI Fluency: Framework & Foundations (Anthropic)
2026	Claude 101 (Anthropic)
2025	Google AI and Technology Academy Graduation Certificate
2025	Bootcamp Finalist Certificate of Achievement (Google AI & Technology Academy)
2025	Google Advanced Data Analytics Certificate
2025	Foundations of Data Science (Google)
2025	The Nuts and Bolts of Machine Learning
2025	Regression Analysis: Simplify Complex Data Relationships (Google)
2025	Go Beyond the Numbers: Translate Data into Insights (Google)
2025	The Power of Statistics (Google)
2025	Get Started with Python (Google)
2025	Foundations of Project Management (Google)
2022	Deutsches Sprachdiplom 1 (Kultusministerkonferenz)